

GIFU AB, Japan

WMO: 476340

Lat: 35.394N

Lon: 136.869E

Elev: 42

StdP: 100.82

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-2.1	-10.0	1.6	3.1	-8.8	1.8	3.5	9.1	4.6	8.3	5.1	1.3	100	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	34.9	25.7	33.2	25.1	32.1	24.7	26.6	31.5	26.1	30.8	25.7	30.1	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.5	28.0	25.0	20.2	27.9	24.2	19.2	27.6	83.4	31.9	81.2	30.5	79.1	30.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.2	DB	-6.0	36.6	1.2	1.1	-6.8	37.4	-7.5	38.0	-8.2	38.7	-9.1	39.5	(4)
(5)			WB	-6.7	27.7	1.0	0.8	-7.5	28.3	-8.1	28.8	-8.6	29.2	-9.4	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	3.5	4.5	8.1	13.9	18.9	22.7	26.8	28.1	24.2	18.2	11.6	6.1	(6)	
	DBStd	8.90	2.39	2.81	3.16	3.32	2.43	2.20	2.42	2.08	2.79	2.86	3.16	3.11	(7)	
	HDD10.0	586	203	154	76	7	0	0	0	0	0	0	19	127	(8)	
	HDD18.3	1944	461	386	318	137	22	1	0	0	1	37	201	381	(9)	
	CDD10.0	2638	0	2	17	123	277	382	521	561	426	255	68	5	(10)	
	CDD18.3	952	0	0	0	3	41	133	263	303	176	34	0	0	(11)	
	CDH23.3	8432	0	0	0	31	283	814	2566	3301	1298	139	0	0	(12)	
	CDH26.7	3126	0	0	0	2	50	194	1040	1433	393	13	0	0	(13)	
(14) Wind	WSAvg	2.2	2.3	2.6	2.8	2.7	2.3	2.1	2.0	2.1	2.0	1.9	1.8	2.1	(14)	
(15) Precipitation	PrecAvg	1776	74	85	125	141	177	210	235	158	247	145	100	78	(15)	
	PrecMax	2199	166	163	230	242	342	374	525	358	509	399	252	127	(16)	
	PrecMin	1188	29	40	40	17	65	80	49	26	61	11	12	14	(17)	
	PrecStd	281	38	33	47	63	65	73	120	89	117	76	58	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	16.2	20.1	26.2	30.1	32.2	36.2	37.0	33.8	28.1	22.1	17.0	(19)	
		MCWB	7.6	10.7	12.1	15.9	18.5	23.6	25.8	25.7	25.1	20.7	15.3	12.3	(20)	
	2%	DB	11.1	13.8	18.0	24.0	28.0	30.2	34.8	35.1	32.0	26.2	20.2	14.8	(21)	
		MCWB	6.3	8.4	10.6	15.5	18.1	22.0	25.5	25.7	24.0	19.4	14.2	9.9	(22)	
	5%	DB	9.9	11.9	16.1	22.0	26.2	29.0	33.1	34.0	30.2	24.9	18.9	13.0	(23)	
		MCWB	5.8	6.9	10.0	14.2	17.3	21.5	25.0	25.6	23.4	18.9	13.5	8.7	(24)	
	10%	DB	8.2	10.1	14.2	20.2	24.9	27.8	31.9	32.9	29.0	23.2	17.1	11.8	(25)	
		MCWB	4.5	5.7	9.0	13.6	16.9	21.0	24.7	25.2	22.7	17.7	13.1	7.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	12.1	14.9	19.8	22.0	25.6	27.0	27.7	26.5	23.5	17.7	13.5	(27)	
		MCDB	11.0	15.1	16.6	21.9	25.0	28.9	32.9	32.5	29.9	25.6	19.6	15.2	(28)	
	2%	WB	7.4	9.7	13.0	17.5	20.4	24.4	26.2	26.9	25.7	21.6	16.2	11.2	(29)	
		MCDB	9.6	11.9	15.6	21.1	24.2	27.7	31.7	31.9	29.1	24.3	18.2	13.5	(30)	
	5%	WB	6.1	7.9	11.5	16.2	19.2	23.5	25.8	26.4	24.9	20.2	15.1	9.6	(31)	
		MCDB	8.6	10.1	14.7	20.0	23.9	26.6	30.9	31.1	28.3	23.2	17.6	12.0	(32)	
	10%	WB	5.0	6.2	9.9	15.0	18.2	22.5	25.3	26.0	24.2	18.9	13.9	8.2	(33)	
		MCDB	7.8	9.5	13.1	18.9	22.7	25.9	30.0	30.4	27.5	22.2	16.4	10.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.7	10.2	10.6	10.2	8.2	7.8	8.5	8.0	8.9	9.5	8.9	(35)	
		MCDBR	10.6	11.8	12.8	13.5	13.2	10.5	10.2	10.2	9.4	10.1	11.0	10.4	(36)	
	5% WB	MCWBR	7.1	7.7	7.8	6.9	6.0	4.3	3.6	3.4	3.8	4.9	6.3	6.7	(37)	
		MCWBR	9.2	9.5	9.8	9.6	9.8	7.6	8.7	8.6	7.5	7.7	8.5	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.393	0.432	0.501	0.542	0.550	0.578	0.565	0.545	0.494	0.456	0.415	0.392	(40)	
		taud	2.137	2.047	1.903	1.858	1.903	1.906	1.995	2.042	2.150	2.170	2.189	2.165	(41)	
	Ebn at Noon	Ebn at Noon	775	789	766	759	759	735	742	748	767	759	748	748	(42)	
		Edh at Noon	119	146	184	202	195	195	177	166	142	128	113	110	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.25	2.98	3.93	4.74	5.23	4.75	4.83	4.97	3.99	3.28	2.48	1.99	(44)	
	RadStd	RadStd	0.16	0.23	0.29	0.44	0.57	0.42	0.69	0.47	0.36	0.31	0.21	0.17	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page